



MATLAB Drive > dsp5_1.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;           % High-pass cutoff frequency (Hz)
7 fs = 8000;           % Sampling frequency (Hz)
8 N = 50;              % Filter order
9
10 wc = fp/(fs/2); % Normalized cutoff (0-1)
11
12 %% --- FIR High Pass using Rectangular Window ---
13 b_rect = fir1(N, wc, 'high', rectwin(N+1));
14
15 disp('--- High-Pass Coefficients (Rectangular Window)');
16 disp(b_rect);
17
18 %% --- FIR High Pass using Hamming Window ---
19 b_hamm = fir1(N, wc, 'high', hamming(N+1));
20
21 disp('--- High-Pass Coefficients (Hamming Window)');
22 disp(b_hamm);
23
24 %% --- FIR High Pass using Hanning Window ---
25 b_hann = fir1(N, wc, 'high', hanning(N+1));
26
27 disp('--- High-Pass Coefficients (Hanning Window)');
28 disp(b_hann);
29
30 %% --- Frequency Response Plots ---
31 figure;
32 freqz(b_rect, 1);
33 title('High-pass FIR using Rectangular Window');
34
35 figure;
36 freqz(b_hamm, 1);
37 title('High-pass FIR using Hamming Window');
38
39 figure;
40 freqz(b_hann, 1);
41 title('High-pass FIR using Hanning Window');
```

Figure 1: dsp5_1

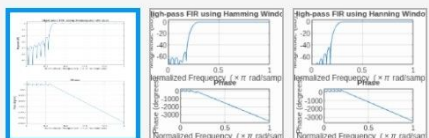
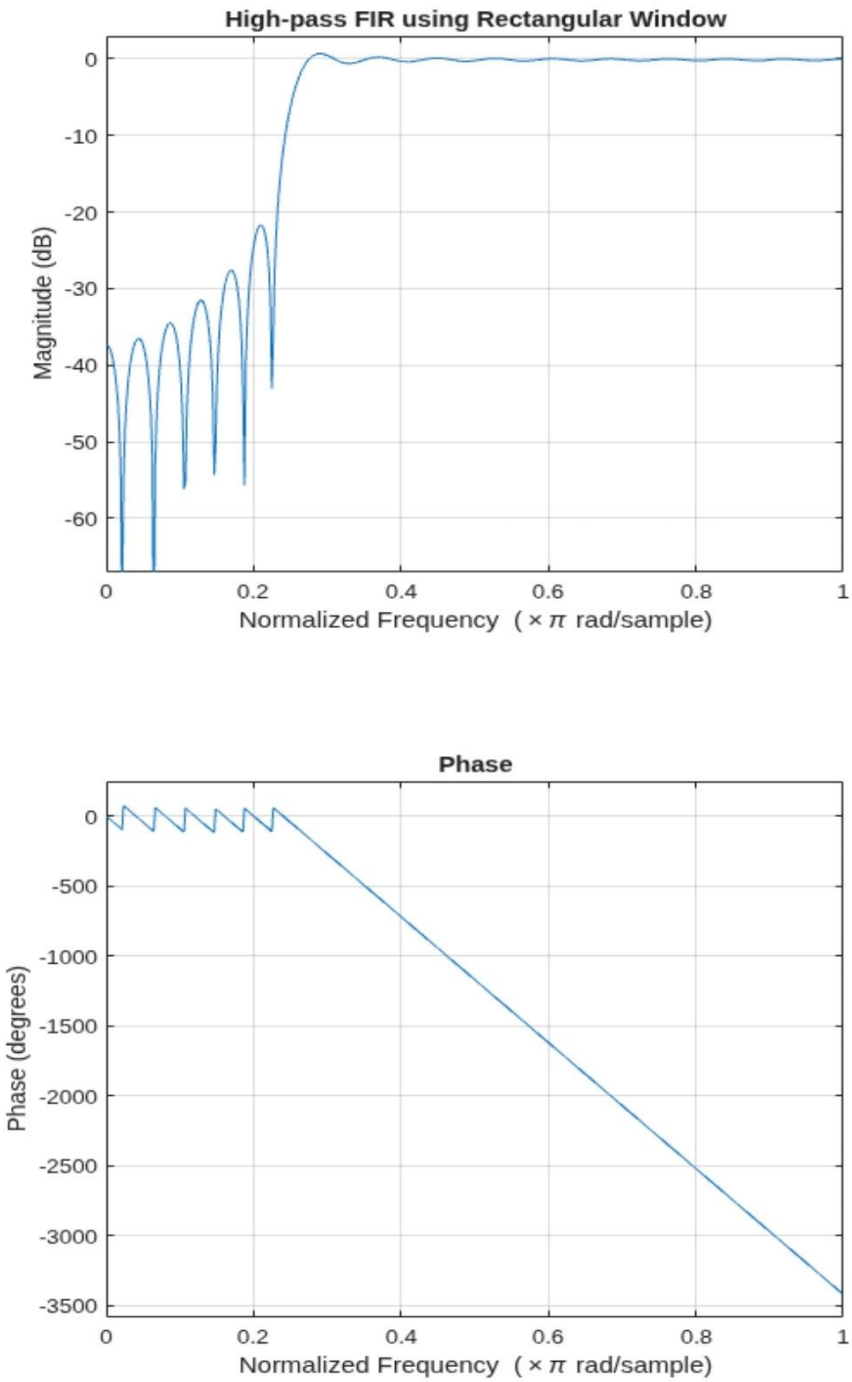


Figure 2: dsp5_1

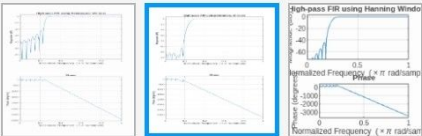
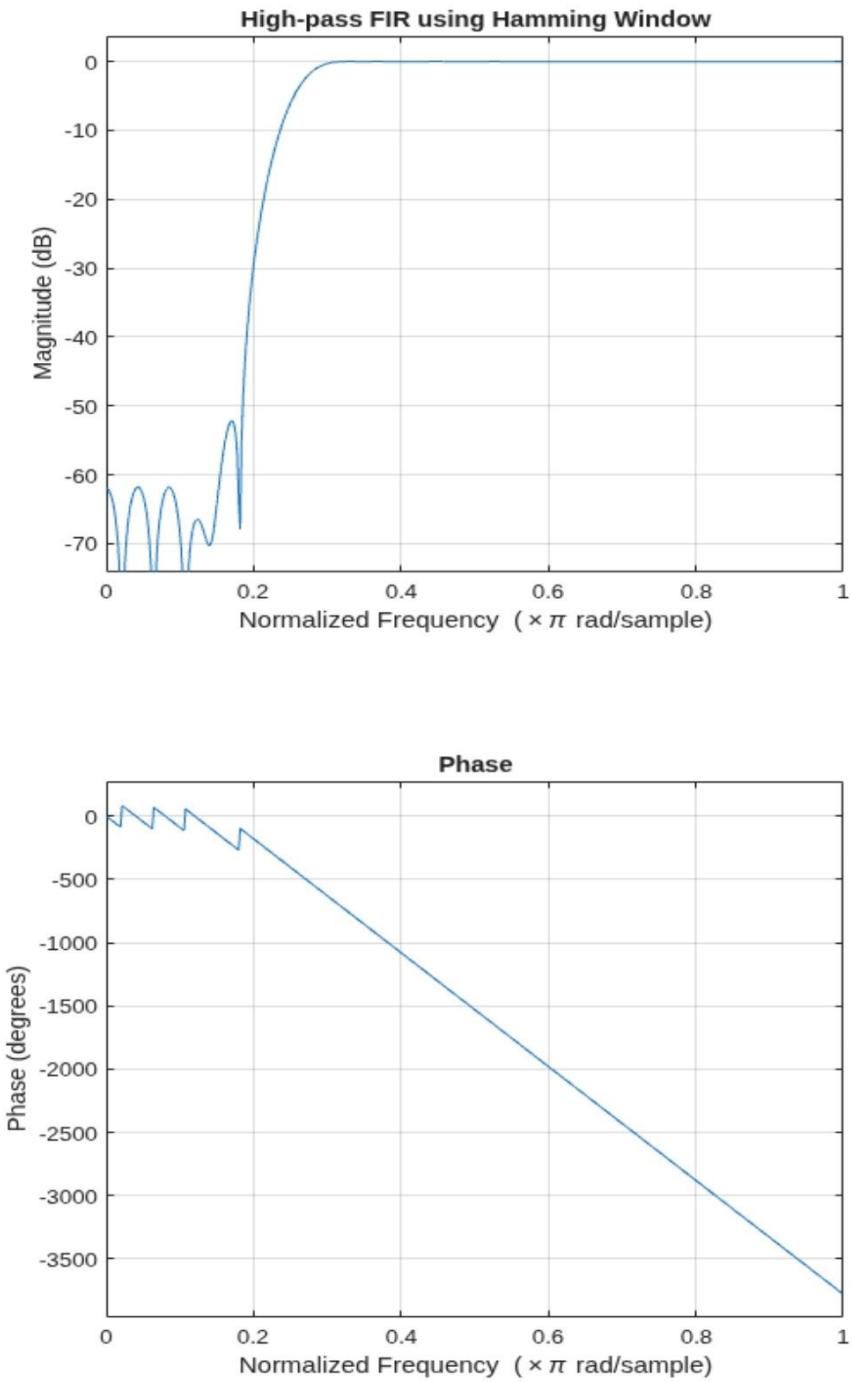
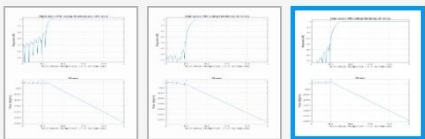
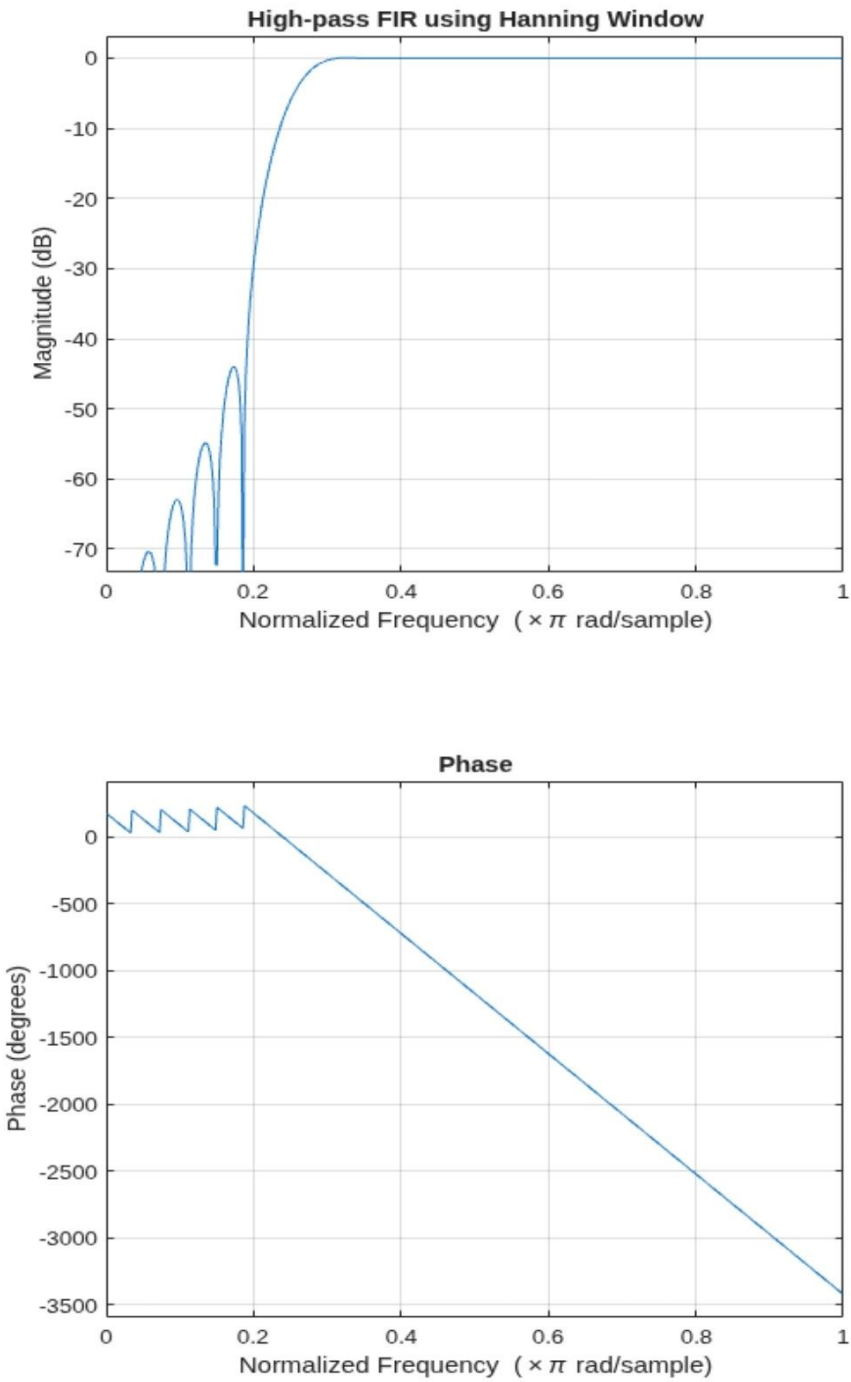


Figure 3: dsp5_1

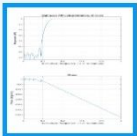
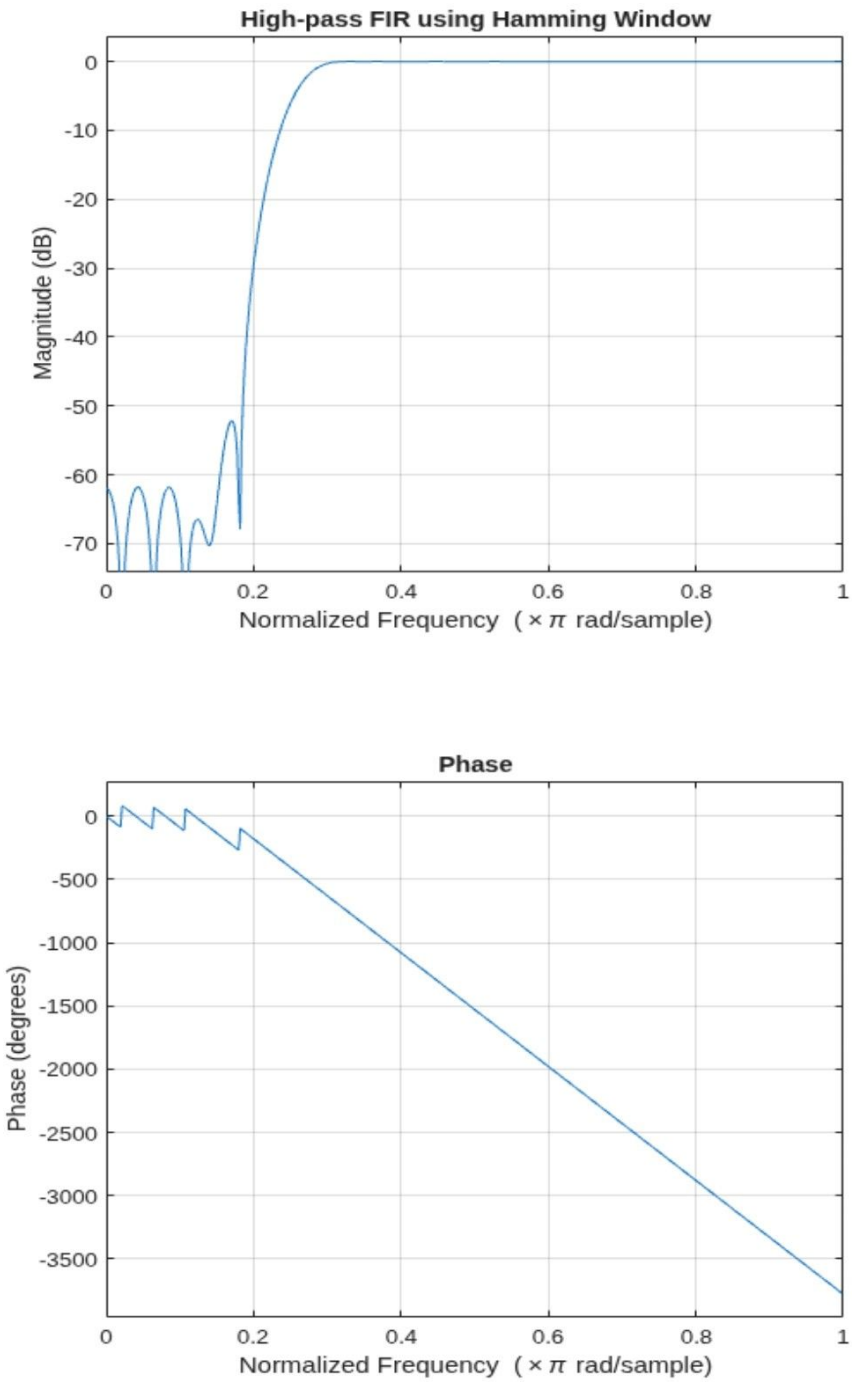




MATLAB Drive > dsp5_2.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;           % High-pass cutoff frequency (Hz)
7 fs = 8000;           % Sampling frequency (Hz)
8 N = 50;              % Filter order
9
10 wc = fp/(fs/2); % Normalized cutoff (0-1)
11
12 %% --- FIR High Pass using Hamming Window ---
13 b_hamm = fir1(N, wc, 'high', hamming(N+1));
14
15 disp('--- High-Pass Coefficients (Hamming Window)');
16 disp(b_hamm);
17
18 %% --- Frequency Response Plot ---
19 figure;
20 freqz(b_hamm, 1);
21 title('High-pass FIR using Hamming Window');
```

Figure 1: dsp5_2





MATLAB Drive > dsp5_3.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;           % High-pass cutoff frequency (
7 fs = 8000;           % Sampling frequency (Hz)
8 N = 50;              % Filter order
9
10 wc = fp/(fs/2); % Normalized cutoff (0-1)
11
12 %% --- FIR High Pass using Hanning Window ---
13 b_hann = fir1(N, wc, 'high', hanning(N+1));
14
15 disp('--- High-Pass Coefficients (Hanning Wind
16 disp(b_hann);
17
18 %% --- Frequency Response Plot ---
19 figure;
20 freqz(b_hann,1);
21 title('High-pass FIR using Hanning Window');
```

Figure 1: dsp5_3

